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Hypertension and associated factors among patients with diabetes mellitus attending a follow-up clinic in central Ethiopia

Yohannes Mekuria Negussie[1] ✉ & Abel Tezera Abebe[2]

Hypertension and diabetes mellitus are highly prevalent chronic diseases worldwide, and their coexistence presents a growing public health challenge, particularly in low-resource settings like Ethiopia. Thus, this study aimed to assess hypertension and its associated factors in patients with diabetes mellitus attending a follow-up clinic in central Ethiopia. A health facility-based cross-sectional study was conducted. Study participants were selected using a systematic random sampling method. Data were collected through a structured, interviewer-administered questionnaire and patient record reviews. A binary logistic regression model was employed to examine the association between hypertension and the explanatory variables. Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) were calculated to estimate the strength of associations, and statistical significance was determined at a p-value of < 0.05. A total of 379 patients with diabetes mellitus participated in the study. The prevalence of hypertension was 34.6% (95% CI: 29.8–39.4). Being aged 41–60 years (AOR = 2.26, 95% CI: 1.23–4.15), having type 2 diabetes (AOR = 3.83, 95% CI: 1.07–13.65), the presence of comorbidities (AOR = 2.72, 95% CI: 1.59–4.67), and poor medication adherence (AOR = 2.06, 95% CI: 1.12–3.77) were statistically significant factors associated with hypertension among diabetes mellitus patients. More than one-third of patients with diabetes mellitus had hypertension. Older age, type 2 diabetes, the presence of comorbidities, and poor medication adherence were factors associated with hypertension. Regular blood pressure checks, especially for older patients, managing comorbidities, and improving medication adherence through education and simpler regimens are recommended.

Keywords Cardiovascular disease, Diabetes mellitus, Ethiopia, Hypertension

Abbreviations

AHMC	Adama Hospital Medical College
AOR	Adjusted odds ratio
CI	Confidence interval
COR	Crude odds ratio
DM	Diabetes mellitus
IQR	Inter quartile range
OHA	Oral hypoglycemic agent

Hypertension and diabetes mellitus (DM) are among the most common chronic diseases worldwide, and their co-occurrence presents a significant public health challenge, particularly in developing countries like Ethiopia[1–3]. This dual burden exacerbates the risk of life-threatening complications, including cardiovascular diseases, which remain the leading cause of mortality globally[4–6]. The combined impact of these conditions places immense strain on healthcare systems, highlighting the urgent need for integrated prevention and management strategies, especially in resource-limited settings[4,7].

According to the International Diabetes Federation (IDF), approximately 537 million people worldwide had diabetes in 2021, with healthcare costs reaching $966 billion[8]. Cardiovascular complications are the leading cause of death and morbidity among individuals with DM, with adults facing a 2–4 times higher risk of cardiovascular

[1]Department of Medicine, Adama General Hospital and Medical College, Adama, Ethiopia. [2]School of Medicine, Faculty of Medical Sciences, Institute of Health, Jimma University, Jimma, Ethiopia. ✉email: yohannes_mekuria@yahoo.com

disease compared to those without DM, with the risk escalating as glycemic control worsens[9,10]. Hypertension, defined as persistently elevated blood pressure, is a global health challenge affecting about 1.39 billion people and causing over 10.8 million deaths annually[11,12]. It is a major contributor to cardiovascular diseases and is common among people with DM[13]. The coexistence of hypertension and DM significantly increases the risk of macro- and microvascular complications, including coronary artery disease, stroke, nephropathy, and retinopathy[4,14,15].

The prevalence of hypertension among individuals with DM is alarmingly high, ranging from 32 to 82% globally[16,17]. In Ethiopia, a meta-analysis reported a pooled prevalence of 55% among patients with type 2 DM[3], significantly higher than the 19.6% prevalence in the general Ethiopian population[13]. Regional studies have shown varying prevalence rates, with reports ranging from 46.5% in Jimma to 59.5% in Debre Tabor[18,19].

The pathophysiology underlying the frequent co-occurrence of hypertension and DM is complex and multifaceted. Insulin resistance, a hallmark of type 2 DM, is crucial in promoting sodium retention and enhancing sympathetic nervous system activity, both of which contribute to elevated blood pressure[20–22]. In addition, the inappropriate activation of the Renin-Angiotensin-Aldosterone System (RAAS) in DM patients further exacerbates hypertension through multiple mechanisms, including vasoconstriction and fluid retention[23]. Hyperglycemia-induced endothelial dysfunction and dyslipidemia can also aggravate the condition by promoting vascular inflammation, oxidative stress, and arterial stiffness[4,20].

Several factors influence the development of hypertension among DM patients, including sociodemographic, clinical, and behavioral characteristics. Older age, male sex, obesity, longer DM duration, and poor glycemic control are well-documented risk factors[24,25]. Lifestyle factors such as physical inactivity, excessive salt intake, and substance use (e.g., alcohol and smoking) further exacerbate the risk of hypertension among diabetic individuals[26].

Limited access to healthcare services, inadequate awareness, and resource constraints pose significant challenges to effective hypertension management[3,27–29]. Despite growing research on hypertension and DM, there is limited information about the prevalence and factors associated with hypertension among patients with DM in Ethiopia, particularly in the study area. Moreover, Ethiopia's three-tier healthcare system, anchored by the Health Extension Program, presents both opportunities and challenges for managing chronic diseases such as hypertension and DM[30]. Thus, this study aimed to address these gaps by offering updated data on the prevalence of hypertension and its associated factors among patients with DM in Adama, Central Ethiopia. The findings will provide valuable insights into the interplay between hypertension and DM, contributing to global efforts to tackle non-communicable diseases in resource-limited settings. Moreover, the insights gained will support healthcare providers and policymakers in designing context-specific interventions to enhance the management of hypertension among DM patients.

Methods

Study design, setting, and period

A health facility-based cross-sectional study was conducted at Adama Hospital Medical College (AHMC) in Ethiopia. AHMC is situated in Adama town, Oromia region, approximately 99 km southeast of Addis Ababa. Founded in 1946, it is among the largest public teaching hospitals in the region, offering preventive and curative services to over six million people and acting as a referral hub for neighboring zones and regions. The study was conducted from June 10 to August 31, 2024.

Population and eligibility criteria

The source population comprised all adults living with DM and receiving follow-up care at AHMC, while the study population included all adult DM patients attending follow-up care during the study period. Patients who had been on follow-up care for at least 6 months during the data collection period were included in the study. However, patients who were critically ill, had a mental illness that impaired participation, were pregnant, or had been newly diagnosed with DM during the study period were excluded.

Sample size determination

The sample size for this study was determined using a formula designed to estimate a single population proportion. The calculation was grounded in the following statistical assumptions:

- A cross-sectional study design to meet the study objectives.
- A hypertension prevalence (P) of 37.4%, derived from a study conducted at Jimma University Medical Center in Southwest Ethiopia[31].
- A confidence level of 95%, corresponding to a standard Z-value of 1.96.
- A margin of error (d) of 0.05.

Thus, by inserting these parameters into the formula:

$$n = \frac{(z\,\alpha/2)^2 * p\,(1-p)}{d^2} = \frac{(1.96)^2 * 0.374\,(1-0.374)}{(0.05)^2} = 360$$

After accounting for a 5% non-response rate, the final sample size was calculated to be 379.

Sampling procedure

The study participants were selected using a systematic random sampling method. The sampling interval was determined by dividing the total population of DM patients registered at the follow-up clinic at AHMC by the calculated sample size. The first participant was selected randomly, and subsequent participants were included at regular intervals.

Variables of the study

Dependent variable

Hypertension (yes/no).

Independent variables

The independent variables included socio-demographic and economic factors (age, sex, place of residence, marital status, educational status, occupational status, and average monthly income), clinical-related characteristics (type of DM, type of treatment, duration of DM, glycemic control family history of DM, family history of hypertension, presence of comorbid disease, medication adherence, and body mass index status) and behavioral-related characteristics (physical activity, smoking, alcohol drinking, and khat chewing).

Operational definition

Hypertension   Patients were considered hypertensive if they had an average Systolic Blood Pressure (SBP) of 140 mmHg or greater, Diastolic Blood Pressure (DBP) of 90 mmHg or greater, or if they were on antihypertensive medication[19,32].

Glycemic control   Good glycemic control was defined as a Hemoglobin A1C (HbA1C) level below 7% or a fasting blood sugar (FBS) level of 130 mg/dl or lower in the absence of documented HbA1C, determined in less than three months. If these conditions were not met, the patient was classified as having poor glycemic control[33,34].

Physical activity   Physical activity was assessed by asking participants how many minutes per day and days per week they spent on activities that involved moderate effort, such as work or sports, which slightly increased breathing or heart rate. Participants were considered physically active if they engaged in such activities for at least 30 min per day on five or more days per week, totaling 150 min or more per week[35,36].

Medication adherence   Medication adherence for antidiabetic medications was assessed based on the percentage of prescribed treatment taken. Adherence was categorized as good if a patient took more than 80% of the prescribed antidiabetic medication, losing no more than two doses within a single year. Poor adherence was defined as losing more than two doses of the prescribed medication in a year[37,38].

Data collection tools and procedure

Data were collected using a structured, interviewer-administered questionnaire and patient record reviews. The questionnaire was adapted from validated scales and published articles[25,31,37–45], and then modified to suit the study's context. A data abstraction format was used to extract relevant information from patient records. The questionnaire covered sections on sociodemographic and economic characteristics, clinical factors, and behavioral characteristics.

Blood pressure (BP) was measured using a pretested standard mercury sphygmomanometer with a properly sized cuff covering two-thirds of the upper arm. Measurements were taken while the patient was seated in a chair with back support at a 90-degree angle. Participants were instructed to rest for at least 5 min before the measurement. If they had consumed caffeinated beverages, a rest period of 30 min was ensured. Any clothing that could interfere with the BP cuff was removed, and measurements were taken on a bare arm with the cuff positioned at heart level. Patients were seated with their legs uncrossed, and they were instructed not to talk during or between measurements. To ensure accuracy, patients were kept relaxed throughout the procedure. Two consecutive BP measurements were taken with a minimum 5-minute interval, and the mean value was recorded.

The questionnaire was initially prepared in English and then translated into the local languages, Amharic and Afan Oromo, with back-translation performed to ensure accuracy. It was then reviewed by subject matter experts and public health professionals to ensure its relevance and contextual appropriateness for the study. Before data collection began, a pretest was conducted on 5% of the sample ($n = 19$) to evaluate clarity and reliability. The survey involved four nurses as data collectors and two public health professionals as supervisors, all of whom received one day of training before the start of the data collection.

Data processing and analysis

Prior to analysis, data processing steps such as cleaning, computing, transforming, coding, recoding, and grouping variables were carried out. Descriptive statistics were used to summarize the study population's characteristics across all variables. For categorical variables, frequency distributions were calculated, while continuous variables were summarized using appropriate numerical measures after assessing data normality with the Shapiro-Wilk test.

Binary logistic regression was used to model the association between hypertension and the independent variables. Initially, bivariable logistic regression analysis was performed to assess crude associations, with a p-value of 0.25 set as the threshold for selecting variables for multivariable analysis. In the multivariable logistic regression analysis, all candidate variables were included to adjust for potential confounders and identify factors significantly associated with hypertension. The strength of the association was expressed as an adjusted odds ratio (AOR) with a 95% confidence interval (CI), and a p-value < 0.05 was used as the threshold for statistical significance. A standard model-building approach was applied to develop the model.

Before analyzing the association between hypertension and the independent variables, the model's suitability was checked using diagnostic tests. The Hosmer-Lemeshow test showed that the model fit the data well (p-value = 0.837). The Nagelkerke R-square was calculated to assess the proportion of variation in hypertension explained by the combined effects of the explanatory variables, showing that 21.1% of the variation was explained by the collective effects of the variables included in the model. Multicollinearity among the variables was assessed using the variance inflation factor (VIF), with values ranging from 1.015 to 1.269, indicating no significant overlap between the variables. All analyses were performed using the Statistical Package for the Social Sciences (SPSS) software, version 27.

Results

Socio-demographic characteristics

A total of 379 DM patients participated in the study. The median age of the participants was 48 years (IQR: 36–59). Among the participants, 203 (53.6%) were male. The majority of participants, 279 (73.6%), resided in Adama. Additionally, 283 (74.3%) were married, and 141 (37.2%) were employed (Table 1).

Clinical and behavioral-related characteristics

Of the participants, 322 (85%) were diagnosed with type 2 DM, and 162 (42.7%) were taking oral hypoglycemic agents (OHAs). Approximately two-thirds, 246 (64.9%), had been diagnosed with DM for less than five years. A total of 346 (91.3%) had good glycemic control, and 313 (82.6%) demonstrated good medication adherence. Additionally, 160 (42.2%) of the participants had body mass index measurements within the normal range, 18 (4.7%) were smokers, and 93 (24.5%) consumed alcohol (Table 1).

Prevalence of hypertension

In this study, the overall prevalence of hypertension among patients with DM was 34.6% (95% CI: 29.8–39.4). The prevalence was 8.8% (95% CI: 1.8–17.4) among type 1 DM patients and 39.1% (95% CI: 33.9–44.7) among type 2 DM patients.

Factors associated with hypertension among patients with diabetes mellitus

Following bivariable logistic regression analysis, age, type of DM, type of treatment, family history of DM, glycemic control, presence of comorbidities, and medication adherence were identified as candidate variables for multivariable logistic regression analysis based on a p-value threshold of < 0.25. After adjusting for potential confounders in the multivariable logistic regression analysis, age, type of DM, presence of comorbidities, and medication adherence were found to be statistically significant factors associated with hypertension at a p-value < 0.05 (Table 2).

The odds of hypertension were twofold greater among DM patients aged 41–60 compared to those aged 40 years or younger (AOR = 2.26, 95% CI: 1.23–4.15). Compared to type 1 DM patients, those with type 2 DM had 3.8 times higher odds of developing hypertension (AOR = 3.83, 95% CI: 1.07–13.65). This study also revealed that DM patients with comorbidities had increased odds of hypertension compared to those without comorbidities (AOR = 2.72, 95% CI: 1.59–4.67). Moreover, in contrast to DM patients with good medication adherence, those with poor adherence had 2.4 times greater odds of hypertension (AOR = 2.06, 95% CI: 1.12–3.77) (Table 2).

Discussion

Hypertension poses a significant global public health challenge. In individuals with DM, its presence not only complicates treatment strategies and escalates healthcare costs but also markedly increases the risk of macrovascular and microvascular complications, as well as mortality[46,47]. Thus, this study assessed the prevalence of hypertension and its associated factors among DM patients in a diabetic care follow-up clinic in Central Ethiopia. The findings revealed that slightly more than one-third of patients with DM had hypertension. Key factors significantly associated with hypertension included age, type of DM, the presence of comorbidities, and adherence to prescribed medications.

The prevalence of hypertension among DM patients in this study was 34.6% (95% CI: 29.8–39.4)0.34. This is comparable to findings from studies conducted in India (37%)[44] and at Jimma University Medical Center, Southwest Ethiopia (37.4%)[31]. However, it is lower than results from a systematic review and meta-analysis in Africa (58.1%)[48] and studies conducted in Cameroon (86.2%)[40], Uganda (61.9%)[41], Libya (85.6%)[42], Nigeria (54.2%)[49], and Northern Sudan (45.6%)[39]. The differences in the prevalence of hypertension among DM patients can be explained by variations in the characteristics of the study populations, differences in how hypertension was diagnosed, access to healthcare, and the quality of treatment. Differences in the lifestyle of participants may further explain the variations. Additionally, the study setting, study period, and sample size could contribute to these differences. Environmental and genetic factors, along with changes in healthcare practices over time, might also contribute to these variations.

The results of the current study revealed a strong association between age and hypertension among DM patients. The odds of hypertension were twice as high among DM patients aged 41–60 compared to those aged 40 or younger. This finding is in agreement with a systematic review and meta-analysis conducted in Africa (58.1%)[48], as well as studies in Nigeria, Northern Sudan[39], Jimma University Medical Center, Southwest Ethiopia[31], Debre Tabor General Hospital, Northwest Ethiopia[19], Southern Ethiopia[25,35], and Kambata Tambaro Zone, Ethiopia[36]. This can be explained by the fact that as people age, the risk of developing hypertension increases due to physiological changes such as decreased elasticity of blood vessels and increased arterial stiffness, which are common with aging[1]. Additionally, older age is often associated with a higher burden of risk

Variables	Categories	Frequency	Percentage
Age (in years)	≤ 40	137	36.1
	41–60	148	39.1
	˃ 60	94	24.8
	Median (IQR): 48 (36–59)		
Sex	Male	203	53.6
	Female	176	46.4
Place of residence	Adama	279	73.6
	Out of Adama	100	26.4
Marital status	Married	283	74.3
	Unmarried	98	25.7
Educational status	No formal education	36	9.5
	Primary (1–8)	92	24.3
	Secondary (9–12)	110	29.0
	College and above	141	37.2
Occupational status	Employed	141	37.2
	Merchant	86	22.7
	Housewife	69	18.2
	Farmer	34	9.0
	Retried	45	11.9
	Others*	4	1.0
Average monthly income (ETB)	< 1500	16	4.2
	1500–3499	63	16.6
	3500–4999	155	40.9
	≥ 5000	145	38.3
Type of DM	Type 1	57	15.0
	Type 2	322	85.0
Type of treatment	OHA	162	42.7
	Insulin	93	24.5
	Both	124	32.7
Duration of DM	< 5 years	246	64.9
	≥ 5 years	133	35.1
Glycemic control	Good	346	91.3
	Poor	33	8.7
Family history of DM	Yes	27	7.1
	No	352	92.9
Family history of hypertension	Yes	29	7.7
	No	350	92.3
Presence of comorbidity	Yes	96	25.3
	No	283	74.7
Medication adherence	Good	313	82.6
	Poor	66	17.4
Body mass index (kg/m^2)	Underweight	42	11.1
	Normal (18.5-24.99)	160	42.2
	Overweight (25–29.99)	104	27.4
	Obese (≥ 30)	73	19.3
Physical activity	Active	89	23.5
	Inactive	290	76.5
Continued			

Variables	Categories	Frequency	Percentage
Smoking	Yes	18	4.7
	No	361	95.3
Alcohol consumption	Yes	93	24.5
	No	286	75.5
Khat chewing	Yes	99	26.1
	No	280	73.9

Table 1. Socio-demographic, clinical and behavioral characteristics of patients with diabetes mellitus patients attending follow-up care at a diabetic follow-up clinic in central Ethiopia, 2024 ($n = 379$). *DM* diabetes mellitus, *OHA* oral hypoglycemic agent. *Student, waitress, daily laborer, driver.

Variables	Category	Hypertension		COR (95% CI)	AOR (95% CI)
		No (%)	Yes (%)		
Age (years)	≤ 40	108 (78.8)	29 (21.2)	1	1
	41–60	77 (52.0)	71 (48.0)	3.43 (2.04–5.79) *	2.26 (1.23–4.15) **
	˃ 60	63 (67.0)	31 (33.0)	1.83 (1.01–3.32) *	0.91 (0.46–1.84)
Type of DM	Type 1	52 (91.2)	5 (8.8)	1	1
	Type 2	196 (60.9)	126 (39.1)	6.69 (2.60-17.19) *	3.83 (1.07–13.65) **
Type of treatment	OHA	104 (64.2)	58 (35.8)	0.72 (0.45–1.17)	0.61 (0.36–1.02)
	Insulin	74 (76.9)	19 (20.4)	0.33 (0.18–0.62) *	0.70 (0.32–1.52)
	Both	70 (56.5)	54 (43.5)	1	1
Family history of DM	No	224 (63.6)	128 (36.4)	1	1
	Yes	24 (88.9)	3 (11.1)	0.22 (0.07–0.74) *	0.83 (0.18–3.74)
Glycemic control	Good	232 (67.1)	114(32.9)	1	1
	Poor	16(48.5)	17 (51.5)	2.16 (1.05–4.44) *	1.80 (0.81-4.00)
Presence of comorbidity	No	203 (71.7)	80 (28.3)	1	1
	Yes	45 (46.9)	51 (53.1)	2.88 (1.79–4.63) *	2.72 (1.59–4.67) **
Medication adherence	Good	216 (69.0)	97 (31.0)	1	1
	Poor	32 (48.5)	34 (51.5)	2.37 (1.38–4.06) *	2.06 (1.12–3.77) **

Table 2. Multivariable logistic regression analysis of factors associated with hypertension among patients with diabetes mellitus patients attending follow-up care at a diabetic follow-up clinic in central Ethiopia, 2024 ($n = 379$). *AOR* adjusted odds ratio, *CI* confidence interval, *COR* crude odds ratio, *DM* diabetes mellitus, *OHA* oral hypoglycemic agent. *Significant at p-value < 0.25 in unadjusted logistic regression analysis, **significant at p-value < 0.05 in adjusted logistic regression analysis, 1 = Reference.

factors such as obesity, physical inactivity, and poor dietary habits, all of which contribute to the development of hypertension[50,51].

In the current study, patients with type 2 DM had higher odds of developing hypertension compared to those with type 1 DM. This finding is concurrent with previous studies[35,52,53]. Type 2 DM is often associated with obesity, insulin resistance, and metabolic syndrome, all of which are known risk factors for hypertension. The presence of these factors in type 2 DM patients leads to increased arterial stiffness, higher sympathetic nervous system activity, and altered renal function, which contribute to higher blood pressure. In contrast, type 1 DM, which typically develops earlier in life and is characterized by insulin deficiency rather than insulin resistance, may not have the same metabolic risk factors that contribute to the development of hypertension[20,52,54].

The presence of comorbidities was another significant factor associated with hypertension in this study. Diabetic patients with comorbidities were found to have increased odds of developing hypertension compared to those without comorbidities. This finding aligns with previous studies conducted in Bangladesh[55], and Ethiopia[35,37]. This association may be attributed to the cumulative impact of multiple conditions on vascular health, systemic inflammation, and metabolic disturbances, which collectively increase the risk of hypertension. Furthermore, polypharmacy, lifestyle limitations, and the systemic stress of managing multiple conditions also contribute to elevated blood pressure[56–59].

Adherence to anti-diabetic medications refers to how closely an individual's medication-taking behavior aligns with medical advice. Good adherence has been linked to improved glycemic control, fewer diabetes-related complications, fewer hospitalizations, lower healthcare costs, and reduced all-cause mortality[60–62]. In this study, DM patients with poor medication adherence were found to have higher odds of developing hypertension compared to those with good adherence. Poor adherence to DM medications can result in suboptimal glycemic control, which increases oxidative stress, endothelial dysfunction, and insulin resistance, increasing the risk of hypertension. It can also worsen DM-related complications like kidney disease and heart problems, which are linked to hypertension. Additionally, it can lead to metabolic syndrome, including dyslipidemia and obesity, and disrupts renal function, which contributes to hypertension[63–65]. These interconnected mechanisms explain why diabetic patients with poor medication adherence are at higher odds of developing hypertension.

Limitations of the study

When interpreting the study's results, it's important to acknowledge some limitations. The cross-sectional nature of the study restricts the ability to determine causal relationships between potential factors and hypertension. Additionally, since the study was conducted in a hospital setting, its findings may not fully reflect the broader community population.

Conclusion

The current study showed that more than one-third of patients with DM had hypertension. Factors significantly associated with hypertension among DM patients included age, type 2 DM, the presence of comorbidities, and poor medication adherence. Based on the findings, it is recommended that healthcare providers prioritize regular blood pressure monitoring for DM patients, especially older individuals and those with type 2 DM. Comprehensive management of comorbidities to reduce their cumulative impact and promote medication adherence through patient education and simplified treatment regimens should also be emphasized. Future research should explore causal pathways and interventions to improve blood pressure control in DM patients.

Data availability

All data and materials are available from the corresponding author without undue reservation.

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Author contributions
YMN conceived the study idea, conducted data extraction, performed statistical analysis, and interpreted the findings. YMN and AA drafted the manuscript, critically reviewed it, and finalized the version for submission. All authors thoroughly read and approved the final manuscript.

Declarations

Competing interests
The authors declare no competing interests.

Ethical approval and consent to participate
Ethical approval for the study was granted by the Institutional Ethical Review Board (IERB) of Adama General Hospital and Medical College (Ref no. AGHMC/5219/9/2016). Participants received comprehensive information about the study's potential risks and benefits, and their voluntary participation or non-participation was confirmed through written Informed consent. Strict measures were taken to uphold privacy and anonymity, safeguarding participants' rights and ensuring confidentiality. All procedures complied with the ethical standards outlined in the Helsinki Declaration.

Additional information
Correspondence and requests for materials should be addressed to Y.M.N.